A Minor Project Synopsis

On

Project ID:BTP2CSE200

Title of Project: Weather Forecasting Using Machine Learning

Submitted in partial fulfillment of the

requirements for the award of the

degree of

BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2025-26 in

Computer Science and Engineering

By: **Palak Ramtri 2410031494, Aryan Kumar 2410031503,**

Sushant Tiwari 2410031232, Anushka Rajput 2410031519,

Akash Singh 2410031663

Under the guidance of **Dr. Udai Bhan Trivedi**

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

IILM UNIVERSITY, GREATER NOIDA,

INDIA

March, 2026

IILM UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA

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Signature of Supervisor: _____

Date: 27/03/26

1. Introduction:

Weather forecasting is an important part of our daily life, as it helps us understand the current weather conditions and also predict what the weather will be like in the future. It plays a major role in helping people plan their activities and make better decisions. Knowing the weather in advance can help us avoid many problems caused by sudden changes in temperature, rainfall, or extreme conditions. For example, if we know that it is going to rain, we can carry an umbrella or postpone outdoor work. In the same way, weather forecasting helps people stay safe and prepared.

Weather information is useful in many different fields. In agriculture, farmers depend heavily on weather forecasts to decide when to plant seeds, irrigate crops, or harvest them. A small mistake in timing due to unexpected weather can lead to major losses, so accurate prediction is very important. In transportation and travel, weather forecasting helps in planning journeys and avoiding delays caused by bad weather. Tourists also check weather conditions before planning trips or vacations. Similarly, organizations and government agencies use weather forecasts to prepare for extreme conditions such as storms, floods, or heat waves. This helps in reducing risks and protecting people and property.

In earlier times, weather forecasting was done manually based on observation, experience, and basic calculations. People used to study clouds, wind patterns, and seasonal changes to guess future weather conditions. However, these methods were time-consuming and not always accurate. With the development of modern technology, computers and advanced systems have replaced traditional methods. Today, machine learning has become a powerful tool for predicting weather conditions more efficiently and accurately.

Machine learning allows computers to learn from past data and identify patterns that humans may not easily notice. It can process large amounts of historical weather data and use it to predict future temperature and conditions. This makes weather forecasting faster, more reliable, and less dependent on manual effort. Among different machine learning techniques, Linear Regression is widely used for predicting continuous values such as temperature. It is simple, easy to understand, and works well with structured datasets.

This project focuses on developing a weather forecasting system using machine learning techniques. The system uses historical temperature data to train a model, which helps in predicting future temperature based on user input. The user can enter a future day value, and the system will predict the expected temperature for that day. This shows how machine learning can be used in a simple and practical way.

In addition to prediction, the system also integrates a weather API to provide real-time weather information. The API allows users to enter the name of any city and instantly get details such as current temperature, humidity, and weather conditions. This makes the system more useful

because it combines both present and future weather information in one place. Users do not need to switch between different applications to get weather updates.

The system is designed to be simple, fast, and user-friendly so that it can be easily used by different types of users. Farmers can use it to plan agricultural activities, travelers can use it to plan their journeys, and general users can use it for daily weather updates. It is also helpful for students who want to understand how machine learning works in real-life applications. The graphical user interface makes it easy for users to interact with the system without needing technical knowledge.

Another important advantage of this system is that it saves time and effort. Instead of manually analyzing data, the system automatically processes the information and provides results quickly. It also helps users make better decisions by giving both current weather details and future predictions.

Overall, this project demonstrates how machine learning can be applied to solve real-world problems in a simple and effective way. By combining prediction techniques with real-time data, the system provides a complete and practical weather forecasting solution that is useful for a wide range of users.

2. Problem Statement:

The weather is always changing, which makes the temperature on the next day hard to predict. Most people use apps to find the current weather, but they are not easy to read.

What we need is something that uses past weather to predict the next day's, while at the same time allowing us to find the current weather without having to open an app or go to a website. This would make life so much easier.

At the heart of this project is a weather system with two important functionalities: to predict the next day's temperature and to show us the current weather, all in one place. It has to be easy to use, even for people who are not tech-savvy, simple but actually useful. And this is the weather system we want to create.

Scope:

This project is about making a weather forecasting system. The weather forecasting system uses machine learning to predict temperatures and provide weather information.

*** The weather forecasting system looks at temperature records to make predictions about temperatures.**

The goal is to forecast temperature and provide weather details such as humidity and current weather conditions. This is for people who want to know the weather without using tools. The people can use the weather forecasting system to know the weather.

The weather forecasting system uses machine learning techniques like Linear Regression and a weather API to gather data. You can enter your city name in the weather forecasting system. The weather forecasting system will show you the weather and future temperature.

This project focuses on temperature prediction and current weather information. The weather forecasting system does not predict wind, air pressure or climate patterns. The weather forecasting systems accuracy depends on the data used to make predictions about the weather.

The project aims to build a simple to use weather forecasting system. The weather forecasting system provides weather information in one place. The weather forecasting system is useful for people who want to know the weather.

The weather forecasting system gives information about the weather now. The weather forecasting system tells you what the weather will be like later.

The weather forecasting system is for people who want to know what the weather is, like. The weather forecasting system helps people know the weather.

3. Literature review:

Weather forecasting has gained significant attention in recent years with the use of machine learning techniques. Earlier, traditional methods were used which were less accurate and time-consuming. With the advancement of technology, researchers are now focusing on data-driven approaches to improve prediction accuracy and efficiency.

1. Machine Learning Based Weather Prediction:

Recent research shows that machine learning models can effectively analyze historical weather data and provide accurate temperature predictions.

(i) Linear Regression Model: **According to Hastie et al. (2009)**, Linear Regression is one of the simplest and most widely used algorithms for predicting continuous variables such as temperature. It uses a mathematical relationship between input and output variables. This model is efficient for structured datasets and provides reliable results.

(ii) Regression-Based Prediction: **Kumar and Singh (2018)** proposed a temperature prediction model using regression techniques. Their model showed that using historical data improves prediction accuracy and reduces errors in forecasting.

(iii) Performance Analysis: Studies have shown that regression models perform well in predicting short-term weather conditions when trained on proper datasets.

2. Data Processing and Feature Representation:

Data preprocessing plays a key role in improving the performance of machine learning models.

(i) Data Cleaning: **According to Sharma et al. (2019)**, preprocessing steps such as removing missing values, normalization, and formatting are necessary to improve model accuracy.

(ii) Feature Selection: **Zhang et al. (2017)** explained that selecting important features from the dataset helps in reducing complexity and improving prediction performance.

(iii) Historical Data Usage: Research shows that using past temperature data helps models learn patterns and trends for better future predictions.

3. Use of APIs in Weather Systems:

Modern weather forecasting systems integrate real-time data using APIs.

(i) Real-Time Data Integration: **According to Khan et al. (2020)**, APIs allow systems to fetch live weather data such as temperature, humidity, and conditions, making the system more practical.

(ii) System Efficiency: Real-time data improves user experience by providing updated and accurate weather information instantly.

(iii) Practical Applications: Combining machine learning with API-based systems results in more effective and user-friendly forecasting applications.

4. Comparison with Traditional Methods:

Earlier methods of weather prediction were based on observation and statistical techniques.

(i) Limitations of Traditional Methods: Traditional forecasting methods were time-consuming and less accurate.

(ii) Advantages of Machine Learning: **According to Mitchell (1997)**, machine learning improves prediction accuracy by learning from data and adapting over time.

(iii) Improved Results: Machine learning-based systems provide faster, more reliable, and efficient weather predictions compared to traditional approaches.

4. Objectives:

The main goal of this project is to create a weather forecasting system that's easy to use and actually works. This system should be able to tell us what the temperature will be in the future by looking at what it was like in the past. It should also give us the weather.

The specific things we want to do in this project are:

* Collect weather data to help us make predictions

* Use something called Linear Regression to predict the temperature

* Let the user tell us when they want to know the temperature and we will predict it

* Get the weather from a special service that gives us this information

* Make a simple and easy to use interface that people will like

* Put the predicted temperature and the current weather together in one system that is easy to use and gives us the weather information we need. The weather forecasting system is what we are trying to build and the weather forecasting system will be useful, for people who want to know the weather.

5. Methodology:

The weather forecasting system works by following some steps. It is made to be simple and easy to use.

* First we collect temperature records. These records are used to create a dataset. The dataset is what we use to teach the machine learning model.

We make sure the data is correct and organized. This helps the model find patterns.

* Next we use Linear Regression on the dataset. The model learns how the day of the year and temperature are related.

The model is taught to predict temperatures based on some information.

The system also uses a weather API. This API provides real-time weather information, like temperature and humidity.

* We created a graphical user interface using Tkinter.

The user can enter their city name to get the weather. They can also enter a date to get the predicted temperature. The system then shows the result in a way. The weather forecasting system provides all this information. It makes the system work. We are talking about the weather forecasting system. The weather forecasting system uses machine learning to make predictions. The weather forecasting system is easy to use. The weather forecasting system provides results.

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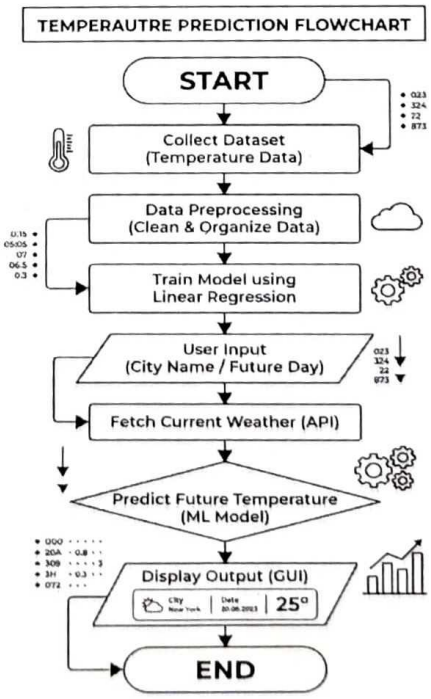


Tools & Technologies:

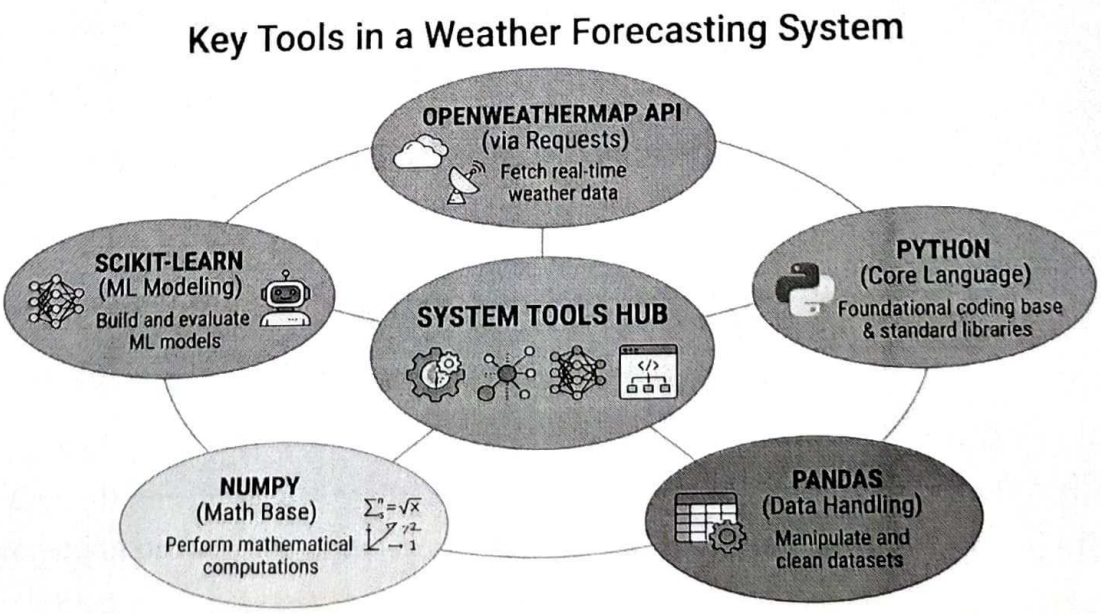
This project is about using engineering ideas and machine learning to make a system that can tell us what the weather will be like. It helps us know what the temperature will be in the future and what the weather is like now by using real time information.

➤ The things we used to make this system work are:

- Python
- Scikit-learn which is used for making models that can predict things
- Pandas to handle all the information
- NumPy to do math problems
- Requests to get information from other places
- Tkinter to make the interface that people use
- Visual Studio Code where we did all the work
- OpenWeatherMap API to get the weather information

All these things work together to make the system work well and give us the right answers. We do this by getting the information teaching the model and then using it to predict what the temperature will be. We also get the weather information from the internet at the same time.

Overall this project uses programming and machine learning to make a system that's easy to use and helps us know what the weather will be, like. The weather forecasting system is made by combining these things. It is really useful. The weather forecasting system is what we are talking about. It is a great tool.



7. Expected Outcomes:

This project is going to create a weather forecasting system that uses machine learning. The weather forecasting system will look at what the temperature was in the past to tell us what the temperature will be in the future. It will also show us what the weather is like using a real-time API.

➤ The things we hope to get from this project are:

- The weather forecasting system will be able to predict the temperature
- It will show us the weather details like temperature and humidity
- The weather forecasting system will have a graphical user interface that's easy to use
- The machine learning model will be integrated properly into the weather forecasting system
- The weather forecasting system will work fast and reliably

The weather forecasting system will help people understand what the weather is like now and what the weather will be like later in a way that's easy to understand. It will also show us how the weather forecasting system can use machine learning in the world.

Overall this project will give us a weather forecasting system application that's easy to use and that combines looking at data and predicting what will happen effectively. The weather forecasting system will be a tool for people who want to know what the weather will be, like.

8. Applications:

This project is really helpful in situations where knowing the weather is important. It helps people make choices based on the weather and the forecast.

➤ This project can be used in many ways, such as:

- Checking the weather every day to plan your day
- Planning what to plant and how to take care of your crops
- Figuring out when to travel and what to do on vacation
- Helping to prevent disasters and giving people warnings when needed
- Teaching students about machine learning in school

The system provides people with information. It shows how machine learning can help solve problems. Overall this project is really helpful, in areas where the weather's crucial. The weather prediction project has practical uses. The weather prediction project helps people in ways.

9. Conclusion and Future Work:

This project is about using machine learning to forecast the weather. It can tell us what the temperature will be like in the future and it can also show the current weather condition. The system is easy to use. It shows people how machine learning can be used to solve real problems.

In the future the weather forecasting project can be made better. We can use data to make it work better. We can also use advanced machine learning to make better predictions. The project can also predict things, like rainfall and wind speed. The weather forecasting project can be made into a website or a phone app so it is easier for people to use the weather forecasting project.

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